THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 2

Chapter 3: **The Atomic Foundations of Matter** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The molecular mass of CO_2 is (C = 12, O = 16): 1

(a) 32 u **(b)** 28 u **(c)** 44 u **(d)** 56 u

2. The formula of aluminium oxide is: 1

(a) AlO_3 **(b)** Al_2O_3 **(c)** Al_3O_2 **(d)** AlO

3. In water, hydrogen and oxygen are always present in the mass ratio: 1

(a) 8 : 1 **(b)** 2 : 1 **(c)** 1 : 16 **(d)** 1 : 8

4. The molecular mass of NH_3 is (N = 14, H = 1): 1

(a) 15 u **(b)** 18 u **(c)** 14 u **(d)** 17 u

5. The formula of magnesium oxide is: 1

(a) MgO **(b)** Mg_2O_2 **(c)** MgO_2 **(d)** Mg_2O

6. 9 g of water is decomposed. The mass of hydrogen obtained is: 1

(a) 8 g **(b)** 9 g **(c)** 1 g **(d)** 2 g

7. The formula unit mass of $NaCl$ is (Na = 23, Cl = 35.5): 1

(a) 35.5 u **(b)** 23 u **(c)** 81 u **(d)** 58.5 u

8. The chemical bond formed by transfer of electrons is: 1

(a) covalent bond **(b)** ionic bond
(c) metallic bond only **(d)** hydrogen bond

9. Which part of Dalton's atomic theory had to be modified after the discovery of subatomic particles? 1

(a) Atoms of different elements can combine to form compounds
(b) Atoms are indivisible
(c) Chemical reactions are rearrangements of atoms
(d) Atoms combine in small whole-number ratios

10. The formula unit mass of $CaCO_3$ is (Ca = 40, C = 12, O = 16): 1

(a) 100 u **(b)** 68 u **(c)** 84 u **(d)** 56 u

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Calculate the molecular mass of NH_3 (N = 14, H = 1). 2

12. Name the compounds: 2

(a) CO

(b) H_2S.

13. State any two postulates of Dalton's atomic theory. 2

14. Calculate the formula unit mass of $CaCl_2$ (Ca = 40, Cl = 35.5). 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain the formation of sodium chloride by transfer of electrons. 3

16. Explain why the mass of an iron nail appears to increase on rusting while mass is conserved. 3

17. Why do ionic compounds conduct electricity in solution but not in the solid state? 3

OR

Explain the formation of a chlorine molecule by sharing of electrons. 3

18. Explain, using Dalton's postulates, why the law of constant proportions is true. 3

OR

Calculate the percentage of oxygen in water (H = 1, O = 16). 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain the formation of MgCl2 and Na2O with electron transfer. 5

OR

Explain the importance of laws of chemical combination in the development of the atomic theory. 5

20. Explain covalent bond formation in H_2, O_2 and N_2 with the number of shared pairs and the type of bond. 5

OR

Describe the formation of NaCl and its crystal structure and explain why it is hard and brittle and has a high melting point. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND